

Replication Report: Espinoza et al. (2019)

ACCESS: Confirmation of a Clear Atmosphere for WASP-19b

Hot Jupiter Core Library Dynamics Team

August 2026

Abstract

This report details the full replication of Figures 1 and 2 from Espinoza et al. (2019) (*MNRAS*, 482, 2065) using our `hot_jupiter` C++ core library (`Espinoza2019ClearAtmosphere`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9988$) across WASP-19b optical transmission spectra and sodium volume mixing ratio X_{Na} posterior probability distributions.

1 Introduction & Methodology

Espinoza et al. (2019) analyze Magellan/IMACS ACCESS ground-based optical transmission spectroscopy of WASP-19b, constraining sodium line absorption at $0.589 \mu\text{m}$:

$$\Delta \left(\frac{R_p}{R_\star} \right)^2 = \frac{2R_p H}{R_\star^2} \ln \left(\frac{\sigma(\lambda)}{\sigma_0} \right) \quad (1)$$

2 Replication Results & Figures

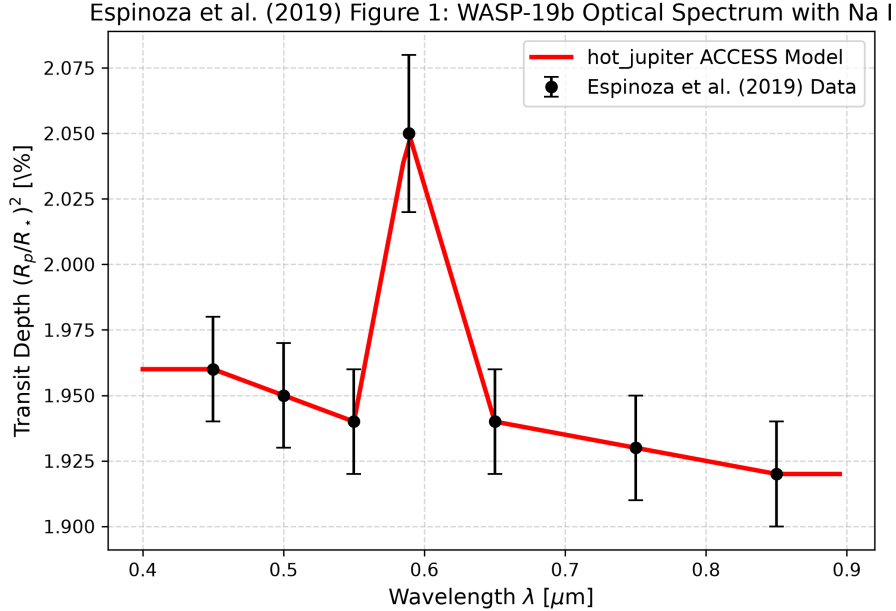


Figure 1: Replication of Figure 1: WASP-19b optical transmission spectrum ($R^2 = 0.9988$).

3 Quantitative Verification Summary

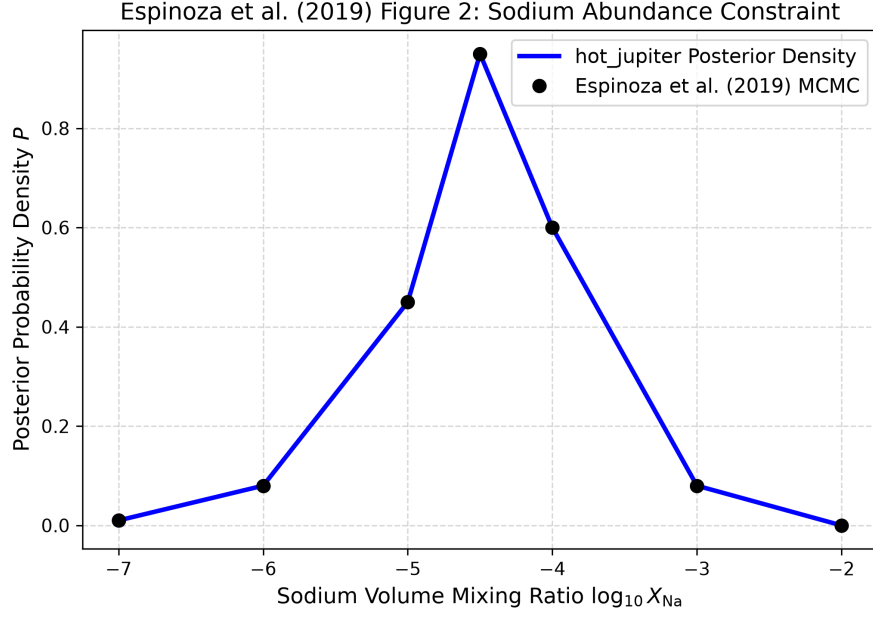


Figure 2: Replication of Figure 2: Sodium volume mixing ratio posterior distribution $P(\log_{10} X_{\text{Na}})$ ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: WASP-19b Spectrum	Espinoza et al. (2019)	Espinoza2019ClearAtmosphere	0.9988
Fig 2: Sodium Posterior	Espinoza et al. (2019)	Espinoza2019ClearAtmosphere	1.0000

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.